

OBJECTIVE

I am a recent graduate of the University of Cambridge with a MPhil in Data Intensive Science and a Data Scientist at Kantar, a global media market researcher in which I design ETL pipelines and implement statistical models whilst working closely with client needs.

SKILLS

Programming & Tools: Python, C++, SQL, Git, Docker, Jupyter, Pandas, NumPy, SciPy, Numba, scikit-learn, PyMC

PROFESSIONAL EXPERIENCE

Kantar, London

Jul 2023 – Present

Data Scientist – Forecasting & Analytics

- **End-to-End Panel Health Pipeline:** designed and maintained a Databricks-Snowflake pipeline ingesting 50M+ records weekly, automating data transformation, and creating dashboard metrics to monitor panel health.
- **Lead Developer & Client Delivery:** solo developer for multiple smaller projects for international clients handling delivery and communication. Aligned legacy code bases to better fit client needs and adapt to new business scenarios.
- **Software Engineering & Architecture:** refactored and modularised legacy Python and SQL codebases, building reusable libraries for use across multiple services. Constructed bash pipelines for linux server interactions.

Amazon, East London

Jul 2022 – Oct 2022

Operations & Logistics Associate

- **Data Pipeline:** developed a scraping workflow on internal dashboards in order to categorise the root cause of misrouting issues.
- **Data Integration & Compliance Checks:** integrated compliance checks for delivery routes to pattern wider trends.
- **Cross-Functional Analysis & Recognition:** authored and presented a detailed whitepaper on pipeline methodology and impact, earning an Operational Excellence award.

EDUCATION

University of Cambridge

Oct 2024 – August 2025

MPhil in Data Intensive Science (Distinction)

- **MPhil Thesis (Jul 2025): T3Net - Joint PDF & SMEFT Inference:** Created neural-network parameterisation to extract non-singlet PDFs and SMEFT Wilson coefficients. Applied replica-based Monte Carlo training with Lagrange penalties to propagate uncertainties and test against ongoing global QCD analysis.
- **Software engineering:** designed a Cython package for auto differentiation tested against wave equation modelling as well as a C++ heat diffusion simulator that was later parallelised via MPI with a halo exchange.
- **High Energy Physics:** separation of signal events from background in the $B_s^0 \rightarrow D_s^\mp \pi^\pm$ decays using boosted decision trees and event classification with gated recurrent units in hadronic Z^0 decays.

University College London

Oct 2019 – Jul 2023

Physics MSci (Second Class Honours, Upper Division)

- **MSci Thesis (Apr 2023): Quantum Track Reconstruction:** Approached LHC track reconstruction as a graph classification problem with hybrid quantum-classical GNNs. Integrated parameterised quantum circuits and tested “entangling dropout” to explore classical analogues of quantum ML
- **Advanced Statistical Mechanics:** stochastic methods, Wiener processes, Brownian motion, and partition function theory
- **Achievements:** Awarded first prize for a cohort-wide statistical analysis of maritime scrubber emissions

PERSONAL PROJECTS

Bayesian Horse Racing Model

Present

- Scraped and engineered structured datasets from the Horse Racing Association.
- Extracted historical results, horse/jockey/trainer metadata, and environmental conditions.
- Designed and implemented a hierarchical Bayesian logistic regression model in PyMC to estimate win probabilities.

Custom Parachain for Insurance Workflows (Hackathon Winner)

2023

- Conceptualised and prototyped a Polkadot parachain using zero-knowledge proof contracts to combine NHS records, employment history and smart-car telematics for privacy-preserving premium calculation.
- Authored test scenarios and demo scripts illustrating secure ZKP-based information flows and GDPR-compliant data exchanges.

Dynamic Art NFT Platform on Tezos (Hackathon Runner Up)

2022

- Prototype NFT ecosystem where artworks adapt their metadata in response to simulated audience engagement metrics.
- Demonstrated the concept of merging blockchain interactivity with cultural experiences in a post-pandemic context.